Linux grep Lab - Commands & Scripts Reference

Part A - Basic grep

Exercise 1 - Search File Contents (using payload.sh)

Find the line containing wget:

```
$ grep "wget" payload.sh
```

Find the line containing nc:

```
$ grep "nc" payload.sh
```

Find the line containing chmod:

```
$ grep "chmod" payload.sh
```

Search for WGET ignoring case:

```
$ grep -i "WGET" payload.sh
```

Display the line numbers containing /tmp:

```
$ grep -n "/tmp" payload.sh
```

Count the number of lines containing /tmp:

```
$ grep -c "/tmp" payload.sh
```

Display only the matched word wget:

```
$ grep -o "wget" payload.sh
```

Search for nc as a whole word:

```
$ grep -w "nc" payload.sh
```

Exercise 2 - Compare Files

Search for wget across files:

```
$ grep "wget" payload.sh update.sh cleanup.sh
```

Identify which file contains wget (list matching filenames):

```
$ grep -l "wget" payload.sh update.sh cleanup.sh
```

Identify which files do not contain wget:

```
$ grep -L "wget" payload.sh update.sh cleanup.sh
```

```
$ grep -l "/tmp" payload.sh update.sh cleanup.sh
```

List files that do not contain /tmp:

```
$ grep -L "/tmp" payload.sh update.sh cleanup.sh
```

Search /tmp across multiple files with filename prefix:

```
$ grep "/tmp" payload.sh update.sh cleanup.sh
```

Exercise 3 - Inverse Search (using system.log)

Display all lines that do not contain INFO:

```
$ grep -v "INFO" system.log
```

Display all lines that do not contain ERROR:

```
$ grep -v "ERROR" system.log
```

Display all lines except those containing WARNING:

```
$ grep -v "WARNING" system.log
```

Count the number of non-INFO entries:

```
$ grep -v "INFO" system.log | wc -l
```

Part B - Security Log Analysis

Exercise 4 - Analyze system.log

Find all ERROR entries:

```
$ grep "ERROR" system.log
```

Find all WARNING entries:

```
$ grep "WARNING" system.log
```

Find the FAILED entry:

```
$ grep "FAILED" system.log
```

Count the number of ERROR entries:

```
$ grep -c "ERROR" system.log
```

Display line numbers containing WARNING:

```
$ grep -n "WARNING" system.log
```

```
$ grep -i "error" system.log
```

Determine whether Malware occurs without displaying output:

```
$ grep -q "Malware" system.log
```

Exercise 5 - Context Investigation

Display two lines after each ERROR (-A 2):

```
$ grep -A 2 "ERROR" system.log
```

Display two lines before each ERROR (-B 2):

```
$ grep -B 2 "ERROR" system.log
```

Display one line before and one line after each ERROR (-C 1):

```
$ grep -C 1 "ERROR" system.log
```

Display two lines around FAILED (-C 2):

```
$ grep -C 2 "FAILED" system.log
```

Part C - egrep / grep -E

Exercise 6 - Search Multiple Security Events

Find lines containing ERROR, WARNING, or FAILED:

```
$ grep -E "ERROR|WARNING|FAILED" system.log
```

Exercise 7 - Multiple Suspicious Commands (using payload.sh)

Search for wget, curl, nc, or chmod using extended regex:

```
$ grep -E "wget|curl|nc|chmod" payload.sh
```

Display only matching words as whole words:

```
$ grep -E -o -w "wget|curl|nc|chmod" payload.sh
```

Check which patterns are not found:

```
$ for x in wget curl nc chmod; do grep -qw "$x" payload.sh || echo "$x";
done
```

Count lines containing at least one of these patterns:

```
$ grep -E -c "wget|curl|nc|chmod" payload.sh
```

Search for commands across multiple scripts:

```
$ grep -E "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
```

Display only filenames containing matches:

```
$ grep -l -E "wget|curl|nc|chmod" *.sh
```

Display matching lines with line numbers:

```
$ grep -n -E "wget|curl|nc|chmod" *.sh
```

Count matching lines in each file:

```
$ grep -c -E "wget|curl|nc|chmod" *.sh
```

Part D - patterns.txt

Exercise 9 - Pattern File Investigation

Search payload.sh using patterns.txt:

```
$ grep -f patterns.txt payload.sh
```

Search system.log using patterns.txt:

```
$ grep -f patterns.txt system.log
```

Search config.conf using patterns.txt:

```
$ grep -f patterns.txt config.conf
```

Search all .sh files using patterns.txt:

```
$ grep -f patterns.txt *.sh
```

Display only filenames containing at least one pattern:

```
$ grep -l -f patterns.txt *.sh
```

Display matching lines without printing filenames:

```
$ grep -h -f patterns.txt *
```

Identify which files contain indicators from patterns.txt:

```
$ grep -l -f patterns.txt *
```

Exercise 10 - Beginning and End of Line Anchors

Find filenames beginning with malware:

```
$ grep "^malware" filelist.txt
```

Find filenames beginning with script:

```
$ grep "^script" filelist.txt
```

Find filenames ending in .sh:

```
$ grep "\.sh$" filelist.txt
```

Find filenames ending in .py:

```
$ grep "\.py$" filelist.txt
```

Find filenames ending in .log:

```
$ grep "\.log$" filelist.txt
```

Find filenames beginning with p and ending in .sh:

```
$ grep "^p.*\.sh$" filelist.txt
```

Exercise 11 - Character Classes

Find malware1.sh and malware2.sh using one pattern:

```
$ grep "^malware[12]\.sh$" filelist.txt
```

Find filenames containing a digit:

```
$ grep "[0-9]" filelist.txt
```

Find filenames containing digits 1 or 2:

```
$ grep "[12]" filelist.txt
```

Find filenames beginning with m or p:

```
$ grep "^[mp]" filelist.txt
```

Find filenames beginning with s or u:

```
$ grep "^[su]" filelist.txt
```

Exercise 12 - Negated Character Classes

Find filenames that do not begin with m:

```
$ grep "^[^m]" filelist.txt
```

```
$ grep "^[^mp]" filelist.txt
```

Find malware filenames where character following malware is not a digit:

```
$ grep "^malware[^0-9]" filelist.txt
```

Part F - File Operators

Exercise 13 - File Existence Check (file_exists.sh)

Script Code:

```
#!/usr/bin/bash
if [ -e "$1" ]
then
    echo "File exists"
else
    echo "File does not exist"
fi
```

Commands executed:

```
$ nano file_exists.sh
$ chmod +777 file_exists.sh
$ ./file_exists.sh payload.sh
$ ./file_exists.sh report.txt
$ ./file_exists.sh unknown.sh
```

Exercise 14 - Check File Type (check_type.sh)

Script Code:

```
#!/usr/bin/bash
if [ -f "$1" ]; then
    echo "Regular file"
elif [ -d "$1" ]; then
    echo "Directory"
else
    echo "Neither"
```

Commands executed:

```
$ nano check_type.sh
$ chmod +777 check_type.sh
$ ./check_type.sh payload.sh
$ ./check_type.sh system.log
$ ./check_type.sh quarantine
$ ./check_type.sh backup
$ ./check_type.sh unknown
```

Exercise 15 - Permissions Check (permissions.sh)

Script Code:

```
#!/usr/bin/bash
file="$1"
echo "File: $file"
echo
if [ -r "$file" ]; then
    echo "Readable: Yes"
else
    echo "Readable: No"
fi
if [ -w "$file" ]; then
    echo "Writable: Yes"
else
    echo "Writable: No"
fi
if [ -x "$file" ]; then
    echo "Executable: Yes"
else
    echo "Executable: No"
fi
```

```
$ nano permissions.sh
$ chmod 777 permissions.sh
$ ./permissions.sh payload.sh
$ ./permissions.sh update.sh
$ ./permissions.sh report.txt
$ ./permissions.sh config.conf
```

Exercise 16 - Empty vs Non-Empty File Check (data_check.sh)

Script Code:

```
#!/usr/bin/bash
if [ -s "$1" ]
then
    echo "File contains data"
else
    echo "File is empty"
fi
```

Commands executed:

```
$ nano data_check.sh
$ chmod 777 data_check.sh
$ ./data_check.sh payload.sh
$ ./data_check.sh empty.txt
```

Part G - File Validation Using [ ]

Exercise 17 - File Validation (validate.sh)

Script Code:

```
#!/usr/bin/bash
file="$1"
if [ ! -e "$file" ]
then
    echo "File does not exist"
```

```
then
    echo "Not a regular file"
elif [ ! -r "$file" ]
then
    echo "File is not readable"
elif [ ! -s "$file" ]
then
    echo "File is empty"
else
    echo "File exists"
    echo "Regular file"
    echo "File is readable"
    echo "File is non-empty"
fi
```

Commands executed:

```
$ nano validate.sh
$ chmod +x validate.sh
$ ./validate.sh payload.sh
```

Exercise 18 - Validation Combined with grep (check.sh)

Script Code:

```
#!/usr/bin/bash
file="$1"
if [ ! -e "$file" ]
then
    echo "File does not exist"
elif [ ! -f "$file" ]
then
    echo "Not a regular file"
elif [ ! -r "$file" ]
then
```

```
elif [ ! -s "$file" ]
then
    echo "File is empty"
elif grep -q "wget" "$file"; then
    echo "Selected indicator detected - investigate further."
else
    echo "Selected indicator not detected."
fi
```

Commands executed:

```
$ nano check.sh
$ chmod +x check.sh
$ ./check.sh payload.sh
$ ./check.sh update.sh
```

Part H - File Tests & Regular Expressions Using [[ ]]

Exercise 19 - Double Bracket Validation (validate_double.sh)

Script Code:

```
#!/usr/bin/bash
file="$1"
if [[ ! -e "$file" ]]; then
    echo "File does not exist"
elif [[ ! -f "$file" ]]; then
    echo "Not a regular file"
elif [[ ! -r "$file" ]]; then
    echo "File is not readable"
elif [[ ! -s "$file" ]]; then
    echo "File is empty"
else
    echo "File exists"
    echo "Regular file"
```

```
    echo "File is non-empty"
fi
```

Commands executed:

```
$ nano validate_double.sh
$ chmod +x validate_double.sh
$ ./validate_double.sh payload.sh
```

Exercise 20 - Regex Extension Matching (extension.sh)

Script Code:

```
#!/usr/bin/bash
filename="$1"
if [[ "$filename" =~ \.sh$ ]]; then
    echo "Shell script identified for inspection"
else
    echo "File does not have a .sh extension"
fi
```

Commands executed:

```
$ nano extension.sh
$ chmod +x extension.sh
$ ./extension.sh payload.sh
$ ./extension.sh scanner.py
$ ./extension.sh report.txt
```

Exercise 21 - Multiple Extensions Regex Matching

Script Code:

```
#!/usr/bin/bash
filename="$1"
if [[ "$filename" =~ \.(sh|py|pl)$ ]]
then
    echo "Script file - inspect if required"
else
```

```
fi
```

Commands executed:

```
$ nano extension.sh
$ ./extension.sh payload.sh
$ ./extension.sh scanner.py
$ ./extension.sh report.txt
$ ./extension.sh config.conf
```

Part I - grep + File Operators + Advanced Logic

Exercise 22 - Executable File Content Analysis (executable_check.sh)

Script Code:

```
#!/usr/bin/bash
file="$1"
if [ ! -e "$file" ]
then
    echo "File does not exist"
elif [ ! -f "$file" ]
then
    echo "Not a regular file"
elif [ ! -x "$file" ]
then
    echo "File is not executable"
elif grep -q "wget" "$file"
then
    echo "wget found"
else
    echo "wget not found"
fi
```

Commands executed:

```
$ nano executable_check.sh
```

```
$ ./executable_check.sh payload.sh
$ ./executable_check.sh update.sh
$ ./executable_check.sh report.txt
```

Exercise 23 - egrep + File Operators (security_check.sh)

Script Code:

```
#!/usr/bin/bash
file="$1"
if [ ! -e "$file" ]
then
    echo "File does not exist"
elif [ ! -f "$file" ]
then
    echo "Not a regular file"
elif [ ! -r "$file" ]
then
    echo "File is not readable"
elif [ ! -s "$file" ]
then
    echo "File is empty"
elif grep -Eq "wget curl nc chmod" "$file"
then
    echo "One or more selected indicators detected"
else
    echo "Selected indicators not detected"
fi
```

Commands executed:

```
$ nano security_check.sh
$ chmod +x security_check.sh
$ ./security_check.sh payload.sh
$ ./security_check.sh update.sh
```

Exercise 24 - Sensitive Information Search

Verify file existence, regular type, readability, and non-empty status:

```
$ [ -e config.conf ] && echo "Exists"
$ [ -f config.conf ] && echo "Regular file"
$ [ -r config.conf ] && echo "Readable"
$ [ -s config.conf ] && echo "Non-empty"
```

Search for password or username patterns in config.conf:

```
$ grep -E "password username" config.conf
```

Part K - Suspicious File Triage Script

Exercise 25 - Comprehensive File Triage (file_check.sh)

Script Code:

```
#!/usr/bin/bash
read -p "Enter filename: " filename
echo
echo "----- FILE INVESTIGATION -----"
echo

if [[ -e "$filename" ]]
then
    exists="Yes"
else
    exists="No"
fi

if [[ -f "$filename" ]]; then
    regular="Yes"
else
    regular="No"
```

```
if [[ -d "$filename" ]]
then
    directory="Yes"
else
    directory="No"
fi

if [[ -r "$filename" ]]; then
    readable="Yes"
else
    readable="No"
fi

if [[ -w "$filename" ]]
then
    writable="Yes"
else
    writable="No"
fi

if [[ -x "$filename" ]]
then
    executable="Yes"
else
    executable="No"
fi

if [[ -s "$filename" ]]
then
    nonempty="Yes"
```

```
        nonempty="No"
    fi

    if [[ "$filename" =~ \.sh$ ]]
    then
        shellscript="Yes"
    else
        shellscript="No"
    fi

    echo "Exists : $exists"
    echo "Regular File: $regular"
    echo "Directory : $directory"
    echo "Readable : $readable"
    echo "Writable : $writable"
    echo "Executable : $executable"
    echo "Non-empty: $nonempty"
    echo "Shell Script: $shellscript"
    echo

    if [[ -f "$filename" && -r "$filename" && -s "$filename" ]]
    then
        grep -q "wget" "$filename"
        if [[ $? -eq 0 ]]; then
            echo "wget : Found"
        else
            echo "wget : Not Found"
        fi

        grep -Eq "wget curl nc chmod" "$filename"
        if [[ $? -eq 0 ]]
```

```
            echo "Selected indicators: Found"
            echo
            echo "Further inspection recommended"
        else
            echo "Selected indicators: Not Found"
        fi
    else
        echo "Selected indicators: Not Found"
        echo "Content analysis not performed"
    fi
```

Execution command:

```
$ nano file_check.sh
$ chmod +x file_check.sh
$ ./file_check.sh payload.sh
```